

What to Do Next in WEKA

A short tutorial for comparing CGPA classification models

Files to use

- `student_performance_primary_clean.csv`: includes previous SGPA.
- `student_performance_behavior_clean.csv`: excludes previous SGPA.

Work with one file at a time. Do not combine their results into the same model ranking.

1

Open the primary dataset

Open **WEKA GUI Chooser**, select **Explorer**, and go to the **Preprocess** tab. Click **Open file** and load the primary CSV.

Check that WEKA shows 878 instances. In the class selector, choose `current_cgpa_band`. It should be the last column.

2

Save an ARFF copy

In the Preprocess tab, click **Save** and save the loaded data as an ARFF file. Use a clear name such as `primary.arff`. ARFF preserves WEKA's attribute types and is easier to reuse.

3

Set the evaluation method

Open the **Classify** tab. Under **Test options**, select:

- Cross-validation
- 10 folds
- Random seed 1

Open **More options** and enable detailed per-class statistics, the confusion matrix, and prediction output when available.

Note: Never use "Use training set" for the accuracy reported in the project.

4

Run the baseline models

First run **ZeroR**. This gives the majority-class baseline. Then run **OneR** as a simple one-feature baseline. Save each result from the result list using names such as `Primary-ZeroR` and `Primary-OneR`.

5

Run the main classifiers

Run the following classifiers using the same 10-fold cross-validation:

- **J48**: Trees → J48
- **Logistic**: Functions → Logistic
- **RandomForest**: Trees → RandomForest
- **SMO**: Functions → SMO

Start with their default settings. If a classifier has trouble with missing values, choose **Meta** → **FilteredClassifier**, set its filter to **ReplaceMissingValues**, and set the required algorithm as the base classifier.

6

Record the results

Create one comparison table and copy these values from every WEKA result:

Model	Accuracy	Macro-F1	Weighted F1	Kappa	Time
ZeroR					
OneR					
J48					
Logistic					
RandomForest					
SMO					

WEKA shows the F1-score for each class. Calculate macro-F1 by averaging the four class F1-scores. Do not confuse this with WEKA's weighted average F1.

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Repeat with the behaviour dataset

Return to the Preprocess tab and load `student_performance_behavior_clean.csv`. Confirm that the class is again `current_cgpa_band`, save an ARFF copy, and repeat Steps 3–6.

Make a separate results table titled **Behaviour-focused model results**. This experiment shows how well the non-SGPA variables work by themselves.

8

Choose the best model

Prefer the model with the strongest **macro-F1**, then check its kappa, accuracy, and confusion matrix. A good model should beat ZeroR and should not perform well for only one CGPA category.

If two models are very close, prefer the easier model to explain, such as J48 or Logistic. A weak behaviour-model result is still a valid finding.

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Examine the errors

For the selected model, save the cross-validation predictions. Convert the class labels to numbers:

Class	C0	C1	C2	C3
Index	0	1	2	3

For each prediction calculate:

$$\text{gap} = \text{actual index} - \text{predicted index}.$$

A gap of 0 is correct, $|\text{gap}| = 1$ is a neighbouring-category error, and $|\text{gap}| \geq 2$ is a larger error. Summarize the counts instead of making personal judgments about individual respondents.

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Finish the report

Your final report should contain:

- The two experiment definitions: primary and behaviour-focused.
- A table comparing all classifiers against ZeroR and OneR.
- The confusion matrix of the selected model.
- A short explanation of the most common errors.
- A comparison of performance with and without previous SGPA.
- Limitations: self-reported data, Semester 7 concentration, and no causal claims.

Note: Do not change models repeatedly just to obtain a higher score. Use the same folds and settings for a fair comparison, and report weak or negative findings honestly.

The exact position of a button may vary slightly between WEKA versions, but the tab and classifier names should remain similar.